EXP 5

Create and Manipulate Factor Variables

Aim:

To create and manipulate factor variables using women's height data and a random sample of letters.

Procedure:

1. Open RStudio or R console.
2. Create a vector containing women's heights.
3. Convert the height categories into a factor variable.
4. Generate a random sample of letters and convert it into a factor.
5. Display the factors and their levels.

Code:

```
heights <- c(150, 155, 160, 155, 165, 160, 170)
height_factor <- factor(heights)

print(height_factor)
print(levels(height_factor))

letters_sample <- sample(LETTERS, 10)
letter_factor <- factor(letters_sample)

print(letter_factor)
print(levels(letter_factor))
```

Output:

```
[1] 150 155 160 155 165 160 170
Levels: 150 155 160 165 170
[1] "150" "155" "160" "165" "170"
 [1] E P K C T Q D U O I
Levels: C D E I K O P Q T U
 [1] "C" "D" "E" "I" "K" "O" "P" "Q" "T" "U"
```